Candidate's Name…….

Signature……

Centre number					Personal No.		

Chemistry
P525/2
Paper 2
Jul / Aug, 2026
2hours

For scorer's use Only		
Total weighted score		
Initials		

KAMTEC EXAMINATIONS BOARD

Uganda Advanced Certificate of Education
CHEMISTRY
PAPER 2(PRACTICAL)
TIME: 2 HOURS

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two compulsory** *examination items.*

Responses to these items **must** *be written in the spaces provided in this paper. Use* **blue** *or* **black** *ink. Any work done in pencil except drawings and graph will not be scored. All working must be clearly shown. Graph paper will be provided.*

Mathematical tables and silent non-programmable scientific calculators may be used.

For Scorer's Use Only			
Item No.	**Basis Code**	**Weighted Score**	**Scorer's Initials**
1	**P**		
	M		
	D		
	DI		
	C		
2	**P**		
	S		
	C		
	Cr		
Total Weighted Score			

TURN OVER

Item 1

Hydrogen peroxide is one of the chemicals used in the treatment of waste water as it oxidizes pollutants and reduces unpleasant odors. However, its use in the treatment of the waste water is effective when the peroxide has not expired. A waste water treatment plant is investigating the time when the hydrogen peroxide will expire as they received the hydrogen without the expiry date on its container.

On doing research, they found that they need to determine the order of reaction with respect to the decomposition of hydrogen peroxide which can then be used to determine the expiry date.

The chief chemical analyst from Uganda analytical center has guided that the peroxide will become expired when the original concentration falls below 80%. He further guided that this can be done by adding a fixed volume of acidified solution of FA_2 and fixed volume of distilled water in a conical flask, followed by $2cm^3$ of starch indicator and varying amount of hydrogen peroxide while taking note of the time taken for the blue colour to appear.

The hydrogen peroxide reacts with the acidified solution of potassium iodide in presence of small amount of starch indicator and sodium thiosulphate according to the equation;

$$H_2O_{2(aq)} + 2I^-_{(aq)} + 2H^+_{(aq)} \longrightarrow 2H_2O_{(l)} + I_{2(aq)}$$

As soon as the iodine is produced, it reacts with thiosulphate and is converted back to the iodide ions as below;

$$I_{2(aq)} + 2S_2O_2^{2-}{}_{(aq)} \longrightarrow 2I^-_{(aq)} + S_4O_6^{2-}{}_{(aq)}$$

When the iodine produced in the reaction is in excess of the sodium thiosulphate, a blue colour suddenly appears as the iodine reacts with starch.

However, the newly recruited personnel, a senior four leaver lacked knowledge on how this can be done and has requested you through your chemistry teacher to guide him.

You are provided with the following;

- *$500cm^3$ of distilled water*
- *$25\ cm^3$ of acidified solution of FA1*
- *$50cm^3$ of FA2 which is the sample of hydrogen peroxide used by the plant.*
- *$10cm^3$ of sodium thiosulphate.*
- *Other apparatus used in chemistry.*

©Kampala Teachers Education Consultancy Examinations, 2026

Tasks

a) Design a scientific investigation you will use to guide the personnel. (*Your design should include the Aim, hypothesis, variables, procedure with two mitigated risks)*

Aim

Hypothesis

Variables

For Scorer's Use Only

Basis code	**P**
score	

Procedure

©Kampala Teachers Education Consultancy Examinations, 2026 **Turn over**

For Scorer's Use Only

Basis code	M
score	

b) Carryout the investigation and record your results *(At least five readings).*

©Kampala Teachers Education Consultancy Examinations, 2026 **Turn over**

For Scorer's Use Only

Basis code	D
score	

c) Analyse your results and interpret it.

For Scorer's Use Only

Basis code	DI
score	

d) Make a conclusion about your result and help the personnel know when the peroxide will become ineffective.

For Scorer's Use Only

©Kampala Teachers Education Consultancy Examinations, 2026 **Turn over**

Basis code	C
score	

Item 2

The knowledge of identify and nature of chemical substances is very important for safe handling of chemicals in school in schools. A secondary school in a certain district ordered for a chemical which is suspected to contain **two cations** and **two anions.**

However, during the handling and transportation of the parcel, the label on the container got removed and the school received the container without labels.

The intern laboratory technician tried to do analysis on the substance and observed that the chemical substance and observed that the chemical substance hardly dissolve in water and suggested that there is need to identify the specific ions in the chemical substance such that it can be re-labelled.

He therefore requested the chemistry teacher to do the analysis on the substance who has in turn tasked you to do the analysis on his behalf.

You are provided with;

- *Substance Z which is the sample of the chemical substance.*
- *Common laboratory reagents for identifying ions.*
- *Some laboratory apparatus*

©Kampala Teachers Education Consultancy Examinations, 2026

Tasks

a) Carryout systematic analysis on the chemical substance and record your observations and deductions.

Tests	Observations	Deductions

©Kampala Teachers Education Consultancy Examinations, 2026

For Scorer's Use Only

Basis code	P	Basis code	S	Basis code	C
score		score		score	

©Kampala Teachers Education Consultancy Examinations, 2026 **Turn over**

b) Help identify the specific ions in the substance and re-label the container.

_____

_____

_____

_____

_____

_____

For Scorer's Use Only

Basis code	**Cr**
score	

END

©Kampala Teachers Education Consultancy Examinations, 2026 **Turn over**